

SOS ALERT SYSTEM

MINI PROJECT REPORT

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SOS Alert System

Overview:

The SOS Alert System is an IoT-based project designed to ensure timely emergency communication. Utilizing an ESP32 module and a simple switch, the system sends alert notifications to the Blynk IoT app in real time when the switch is pressed. This provides a quick and effective way to alert others in critical situations. The system uses minimal components, making it affordable and portable.

Components Used:

ESP32 Module: A Wi-Fi-enabled microcontroller responsible for connecting to the internet and sending notifications.

Switch: Acts as the trigger for the SOS alert system, sending a signal to the ESP32 module when pressed.

Blynk IoT App: A platform to receive real-time alerts from the ESP32 module.

Micro-USB Cable: Used for powering the ESP32 and programming it.

Jumper Wires: Connect the switch to the ESP32 module.

Features and Functionality:

Emergency Alert: When the switch is pressed, the ESP32 module immediately sends an alert to the Blynk IoT app.

Real-Time Communication: Alerts are sent in real time to ensure quick responses during emergencies.

Compact and Cost-Effective Design: The system uses minimal components, making it affordable and portable.

CODE:

```
#define BLYNK_TEMPLATE_ID "TMPL3m4yYs9qU"
#define BLYNK_TEMPLATE_NAME "Notification"
#define BLYNK_AUTH_TOKEN "a2qdC4AIDau3w3qPGmI3LcRx9CGBTyTm"
#define BLYNK_PRINT Serial

#ifdef ESP32

#include <WiFi.h>
#include <WiFiClient.h>

char ssid[] = "Noorul";    // type your wifi name
char pass[] = "noorul23"; // type your wifi password
String Message="SOS Alert!";

#include <BlynkSimpleEsp32.h>

#define Push_Button 35
#define wifi_led 33

#endif

#ifdef ESP8266

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>

#define Push_Button D1
#define wifi_led D4

#endif

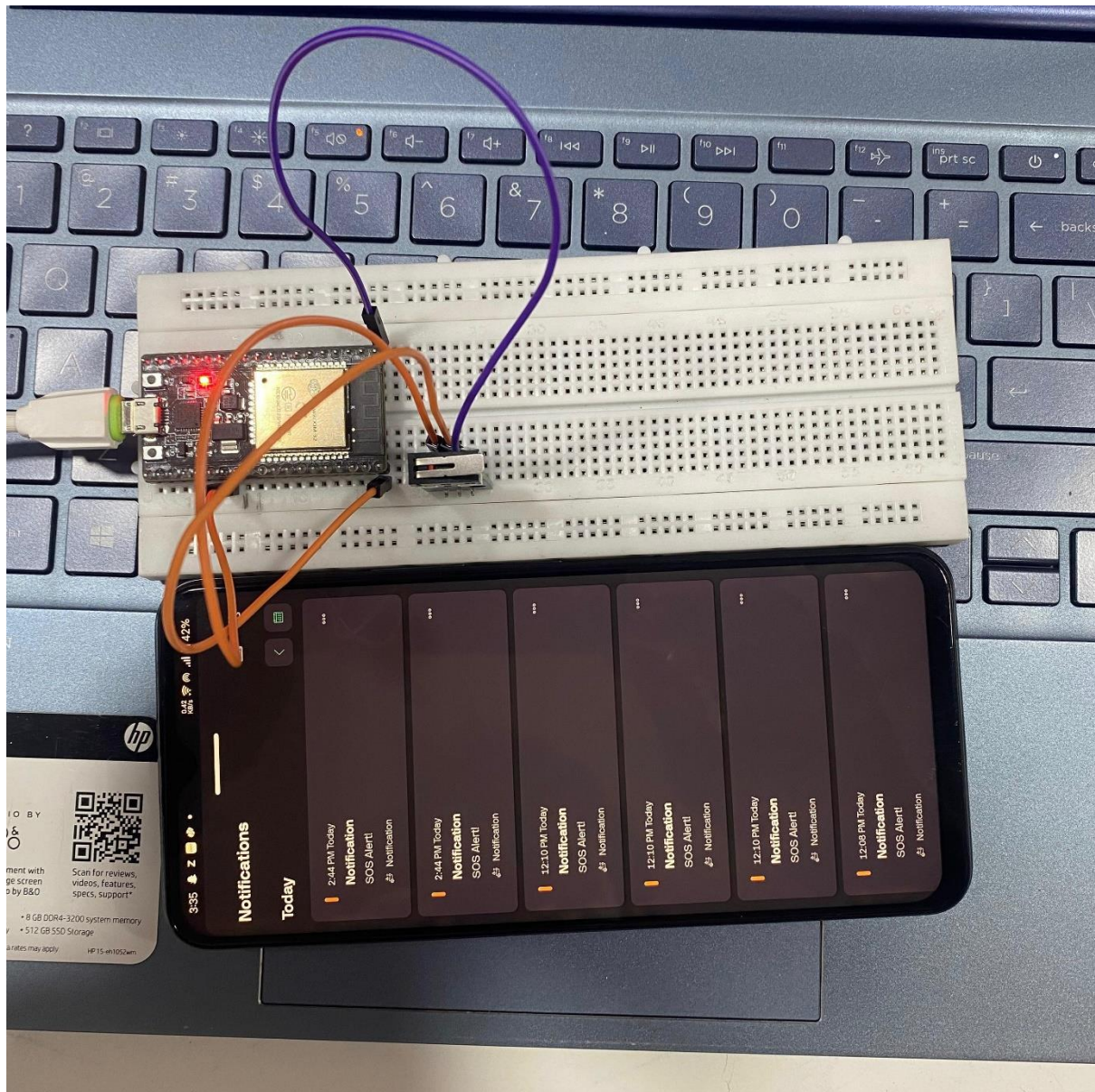
char auth[] = BLYNK_AUTH_TOKEN;
unsigned long old = 0;
unsigned long current = 0;
int interval = 10;
int interval2 = 0;
```

```
void setup() {  
  Serial.begin(115200);  
  Serial.println("\nPlease wait for Blynk Server connection");  
  pinMode(wifi_led, OUTPUT);  
  pinMode(Push_Button, INPUT);  
  WiFi.mode(WIFI_STA);  
  WiFi.begin(ssid, pass);  
  wifi_testing();  
  Blynk.begin(auth, ssid, pass);  
  Blynk.logEvent("notification", Message);  
}
```

```
void loop() {  
  Blynk.run();  
  wifi_testing();  
  current = millis();  
  interval2 = (current - old) / 1000;  
  if (((digitalRead(Push_Button) == LOW) & (interval2 >= 3)) {  
    Serial.println("Blynk notification Sent to Mobile");  
    Blynk.logEvent("notification", Message);  
    while (digitalRead(Push_Button) == LOW)  
      delay(10);  
    current = millis();  
    old = millis();  
    interval2 = 0;  
    if (((digitalRead(Push_Button) == HIGH)) {  
      Serial.println("Blynk notification Sent to Mobile");  
      Blynk.logEvent("notification", Message);  
      while (digitalRead(Push_Button) == LOW)
```

```
    delay(10);  
  }  
}  
  
void wifi_testing() {  
  while (WiFi.status() != WL_CONNECTED) {  
    delay(250);  
    Serial.print(".");  
    digitalWrite(wifi_led, LOW);  
    delay(250);  
    digitalWrite(wifi_led, HIGH);  
    current = millis();  
    if ((current - old) / 1000 > interval)  
      ESP.restart();  
  }  
}
```

Output and Screenshot:



Strengths of the Project:

Quick Response Time: The system ensures that alerts are sent instantly when the switch is pressed.

Simplicity: The design is easy to assemble and requires minimal technical expertise.

Low Cost: The components used are affordable and widely available.

IoT Integration: Real-time notifications via the Blynk IoT app make the system modern and efficient.

Areas for Improvement:

Additional Alert Mechanisms: Adding features like SMS or email notifications for broader reach.

Power Backup: Integrating a rechargeable battery to ensure operation during power outages.

Location Sharing: Incorporating a GPS module to send the user's location with the alert.

Future Enhancements:

Multi-Trigger System: Allowing multiple switches to send alerts from various locations.

Vibration Alerts: Adding a vibration motor for silent alerts.

Smartwatch Integration: Enabling alerts to be sent or received via wearable devices.

Cloud Data Logging: Storing alerts in the cloud for analysis and tracking.

Conclusion:

The SOS Alert System demonstrates how IoT technology can be used to provide efficient and cost-effective solutions for emergency communication. With future upgrades, it can be adapted to cater to diverse needs, making it a highly versatile system.